

systems can be used for various purposes. Applications produce structured, semi-structured and unstructured data, and there are tools and technologies to process the various types of data.

- Separate 'chalk from cheese'. The available data could also contain a lot of junk data that needs to be weeded out. These could be repeatable elements, system generated messages or duplicated data elements.
- Find the 'Holy Grail'. Process the useful data, and refine and enrich it to make the data readily usable and inferential.
- Make it 'sustainable'. Create strong data governance standards and build metadata to easily manage the data lifecycle throughout the organisation.

Data architecture for cloud-native applications

Key architectural considerations for cloud-native applications are:

- Create a data-service combo, where a service is fully handled by a database. This ensures that the service can be managed and scaled independently.
- Separate reads and writes by strategically replicating the data so that applications that only need to access the data can independently handle their needs without any cross-dependencies on data updates.
- De-couple and build asynchronous data components so that producers of data are separated from the consumers, and each can process according to their needs.
- Separate operational data stores from analytical stores, and build data pipelines that transform and load data as per the needs of the organisation.
- Adopt data privacy and security principles by properly masking the data and providing access to users as per their needs.
- Always back up your data and restore it based on your recovery point and recovery time needs.
- Avoid data-centre concentration risks by leveraging the power of the cloud and storing data in multiple areas/zones.
- Adopt modern architectural styles like data-mesh, where data is decentralised and made ready for use by the domain teams.

In summary, creating a data architecture is both a top-down and bottom-up exercise. Business owners, data architects, data science/data analytics teams, development teams and governance/security teams need to work together in a cohesive manner to understand the data needs, and then build a data platform and architecture that meets the needs of the organisation in a sustainable manner. **END** 🐧

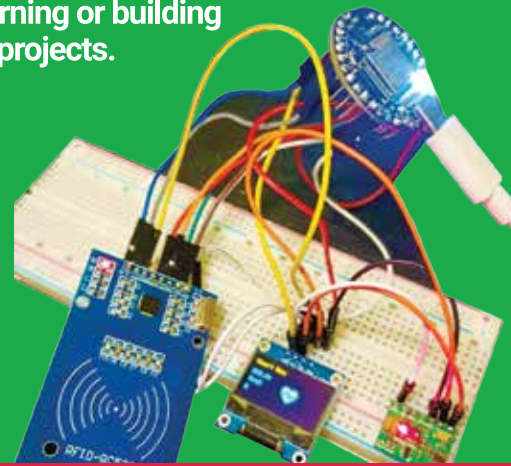
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